



Fig. 10: Dazer lasers: (a) Guardian (b) Defender

With the optional eye-safety module, the device automatically shuts off when the targeted personnel are within the NOHD.

GLARE LA-9/Palco, built around a 250mW green laser, is yet another long-range visual deterrent laser device for hail and warning applications. It is intended to be effective from 0.3km to 4km for ship-to-ship signaling or airborne over-watch applications. This system has an additional feature of automatically shutting off the device if the subject target is within the NOHD. The device is equipped to rapidly detect if a bystander within inside the NOHD; it shuts off the glaring laser output to prevent unintended eye injury.

Saber 203 Grenade Shell Laser Intruder Countermeasure System is a type of laser dazzler that uses a 250mW red laser diode mounted in a hard plastic capsule in the shape of a standard 40mm grenade. It is suitable for being loaded into a M203 grenade launcher and has an effective range of 300m. It is controlled via a box snapped under the launcher, with batteries and a firing switch housed in this box. In emergencies, it can be quickly ejected and replaced with a grenade. Saber 203 dazzlers were used in Somalia in 1995 during Operation United Shield. Fig. 9 shows the photograph of another laser dazzler, which oper-

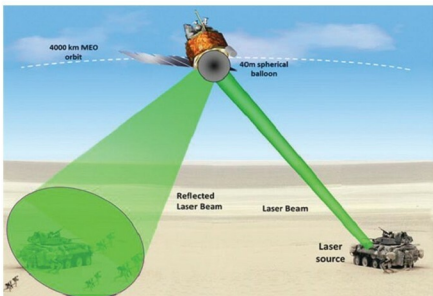


Fig. 11: Laser dazzler for global coverage

ates on green wavelength. Mounted on M240B, it was used during Iraq war.

The Chinese JD-3 laser dazzler and ZM-87 portable laser disruptor are other established systems. JD-3 laser dazzler is reported to be mounted on Chinese Type 98 main battle tank and is coupled with a laser radiation detector, and automatically aims for the enemy's illuminating laser designator, attempting to overwhelm its optical systems or blind the operator.

ZM-87 portable laser disruptor is an electro-optic countermeasure laser device. It can blind enemy troops up to a range of 3km and temporarily blind them at up to 10km range. ZM-87 has been widely-deployed, including its use on naval vessels.

Photonic Disruptor, classified as Threat Assessment Laser Illuminator (TALI), is yet another non-lethal high-power green laser developed by Wicked Lasers, USA, in cooperation with Xtreme Alternative Defence Systems. This tactical laser is equipped with a versatile focus-adjustable collimating lens to compensate for range and power intensity when used to either incapacitate an attacker in close range or safely identify threats

from a distance. TALI-series devices are configured around a 100mW, 532nm laser producing a laser beam with 1.5mrad to 7.5mrad adjustable beam divergence.

Dazer Laser (Fig. 10) from Laser Energetics Inc., USA, is yet another very popular device. It comes in two variants, namely, Guardian that has a range of 1m to 300m (model-dependent) and Defender that has a range of 1m to 2400m (model-dependent). Both variants temporarily impair the vision of the target adversary and succeed in eliminating the threat's ability to see, engage or effectively target the user. This provides the user with a significant advantage over the threat at longer and safer stand-off distances. Both variants are designed to be eye-safe at all ranges beyond 1m and meet the current ANSI safety standard Z136.1.

A vehicle-mounted laser dazzler system designed to control violent crowds and unruly mobs was earlier shown in Fig. 3. Defence Research and Development Organization (DRDO), India, has also developed a short-range hand-held laser dazzler and a vehicle-mounted laser dazzler for crowd-control applications.

*To be continued...*